PSP-MapDiff: Progressively Semantic Prior Information-Guided Online Map Generation From Remote Sensing Imagery via Diffusion Model

Jilong Tian, Peiqin Li, Mengyu Ma, and Hao Chen

***Abstract*—Online map generation from remote sensing (RS) imagery represents a critical yet challenging task in the geographic information system. With the development of deep learning, automatic generation has become a popular research focus. Current mainstream approaches involve two paradigms: domain mapping utilizing generative models and semantic segmentation based on discriminative models. Although these methods provide an effective tool for automatic online map generation, they still face a fundamental dilemma of suffering from poor inferential creativity or low semantic consistency. Therefore, we argue that lacking proper guidance may result in some semantic distortion in the generated maps. To address this issue, we propose PSP-MapDiff, a novel end-to-end network based on a generative denoising diffusion model (DM) incorporating semantic prior information provided by pretrained semantic segmentation networks. In particular, we design a conditional diffusion framework with a denoising starting point from source domain images, effectively driving the denoising process in a promising direction. In addition, the proposed semantic consistency gradient (SCG) and boundary preservation gradient (BPG) modules progressively utilize semantic prior information to enhance cross-domain translation accuracy. Geometric consistency loss is further designed to mitigate distortion and improve robustness. Extensive experiments demonstrate that PSP-MapDiff achieves improvements in visual quality (e.g., +27.8% in LPIPS) and accuracy (e.g., +2.3% in ACC) compared to other state-of-the-art methods, indicating the superiority of the proposed model in maintaining the semantic details of the source domain images.**

***Index Terms*—Diffusion model (DM), online map generation, progressive guidance, remote sensing (RS) images, semantic prior information.**

I. Introduction

THE online map plays a significant role in our modern geospatial applications, including urban planning [1], transportation [2], and emergency management [3]. The advancement of remote sensing (RS) technology has provided unprecedented opportunities for online map generation by offering rich, high-resolution, and timely imagery. Although traditional methods enable making accurate online maps based on professional knowledge, they still rely on manually or semi-manually extracted vector data, which is labor-intensive and time-consuming. This limitation highlights the need for automated and efficient alternatives to quickly generate and update online maps [4]. Over the past decade, the rise of deep learning in image processing has brought about a transformative trend for online map generation [5], enabling conditional image generation in a data-driven manner. With its powerful capability of automatic feature extraction, deep learning has become a necessary complementary tool for online map generation from RS imagery in emergencies [6].

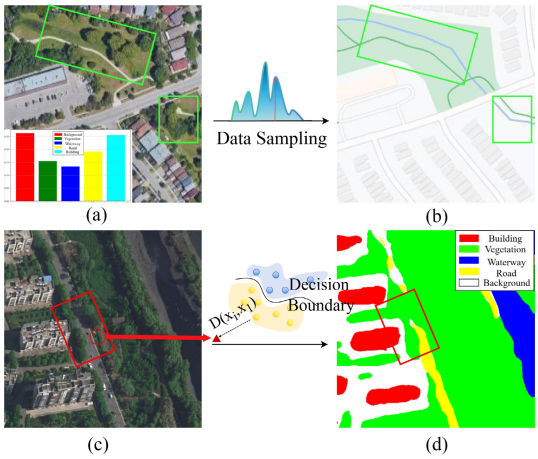


Fig. 1. Online maps generated by different approaches. (a) and (c) RS image in the urban areas. (b) Online map generated by the generative model. The green rectangles illustrate the errors of semantic classification. (d) Segmentation results based on the discriminative, resulting in discontinuity of the road.

Currently, there are two main branches of methods based on deep learning to end-to-end generate online maps from RS imagery automatically, as shown in Fig 1. One is based on generative models, which view online map generation from RS images as a domain mapping task [4]. The other kind of methods employs discriminative models for RS image semantic segmentation [7], relying on various high-quality pixel-level annotations to extract geographic elements, such as buildings [8], roads [9], and edges of waterways [10]. In fact, due to their distinct characteristics, each type of model

Received 15 June 2025; revised 2 October 2025 and 7 November 2025; accepted 8 January 2026. Date of publication 9 February 2026; date of current version 5 March 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 42471403, Grant 42571548, and Grant 42501540. *(Corresponding author: Hao Chen.)*

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Digital Object Identifier 10.1109/TGRS.2026.3662515

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has unique advantages and disadvantages in the online map generation task.

On the one hand, generative models can establish the joint probability distribution between two domains and reflect more of the distribution information of the real data, offering results with creativity and high fidelity [11], [12]. Nevertheless, generative models rely on a latent representation and cannot provide explicit semantic information for each pixel of images, leading to uncontrollable diversity and difficulty in preserving semantic consistency. For example, in Fig. 1, semantic consistency refers to the accurate preservation of semantic categories (e.g., roads, buildings, waterways, and vegetation) in generated online maps. However, the green land area in Fig. 1(a) is not depicted on the corresponding maps shown in Fig. 1(b). Nevertheless, the generated map tiles exhibit good visual quality.

On the other hand, in contrast with generative models, well-trained discriminative models can provide explicit semantic information for each pixel in map generation [13]. However, the performance of the discriminative models usually relies heavily on a large amount of pixelwise annotated training data. Since pixel-level annotations for RS images are always labor-intensive and time-consuming, the discriminative models for map generation usually suffer from either insufficient training data or high costs of manual annotation. Besides, objects in RS images are often affected by occlusion and blurring, which challenge the performance of segmentation and result in inevitable visual discontinuity in these regions [14]. In Fig. 1(c), map producers are more concerned about the accurate generation of roads and buildings, yet the discriminative model can only classify pixels in the red rectangle box as the vegetation class rather than showing the obstructed road. This reflects that the key bottleneck lies in model inference capability for the map generation task.

Therefore, the aforementioned analysis underscores the need for a hybrid approach that leverages the strengths of both types of models. Fu et al. [12] first explore the combination of generative and discriminative models, where the pretrained segmentation network extracts the semantic prior information to guide the process of map generation. Furthermore, Li and Guo [15] propose SPN2DGAN, attempting to introduce semantic and edge labels as prior information guidance to enhance the performance of semantic consistency. These research works employ well-annotated labels as semantic prior information, regularizing the generator to achieve more accurate generation results.

Nevertheless, the online map is abstracted from scenes of reality using human creativity, where there is a remarkable discrepancy between the realistic RS images and abstract maps. The aforementioned models based on adversarial learning try to achieve cross-domain translation in one step, which may suffer from unstable performance facing a large domain gap. While recently emerged diffusion models (DMs) [16], [17] provide a progressive learning strategy, the multistep denoising process achieves a more stable image generation process. Since there is only a relatively slight gap between the adjacent denoising steps, DMs can effectively align the feature distributions by applying fine-tuning [18]. Therefore, we argue that incorporating discriminative models into the progressive denoising process of the DM can obtain more semantically consistent, creative, and stable transformations across the large gap.

Based on the above-mentioned considerations, a novel progressively semantic prior information-guided diffusion framework is proposed in this study. This framework builds on the Brownian motion strategy to achieve an end-to-end translation from RS images to online maps, which can guide the denoising operation in a promising direction. On top of that, the framework further proposes two modules: the semantic consistency gradient (SCG) and the boundary preservation gradient (BPG) guidance modules. First, we employ a segmentation network to extract semantic ground category information from the generated results. It enables the network to generate corresponding semantic content in online maps. In addition, we argue that boundary accuracy is a unique feature for map generation; therefore, we design a boundary loss function to make full use of edge prior information. The two modules compute loss functions that inject gradient guidance into the denoising process. This novel guidance strategy tries to bridge the gap between realistic RS imagery and abstract maps, ensuring global visual quality and local detail fidelity. To verify the effectiveness of our method, we selected three open datasets with significantly different urban scenes from Las Vegas, Cairo, and Rio [12]. The main contributions of this study are as follows.

1) A semantic prior information-guided DM is proposed to generate online maps from RS images. It is a novel framework to incorporate pretrained discriminative models with DMs in online map generation, which preserves inference creativity and semantic consistency.
2) We propose a novel semantic gradient guidance strategy to maintain the semantic consistency and boundary continuity of online maps in the progressive denoise process of the DM. With the help of the progressive guidance of semantic prior information, the proposed model can achieve more semantically consistent, creative, and reliable transformation performance across the large domain gap from realistic RS images to abstract online maps.
3) To assess the effectiveness of the proposed model, extensive experiments have been conducted on three widely used real-world datasets collected from three urban areas with distinct features. The qualitative and quantitative results demonstrate that our method outperforms several state-of-the-art methods in global visual quality and local details.

The structure of this article is organized as follows. The related work of online map generation based on deep learning is shown in Section II. In Section III, we present details of the PSP-MapDiff methodology. The analysis of experimental results is provided in Section IV. Section V makes a comprehensive discussion of our model. Section VI draws the conclusions of this article.

II. Related Works

This section primarily discusses online map generation. We start by introducing the existing generation methods based on

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deep learning. Then, we explore image translation mechanisms of DMs. Finally, we discuss semantic label guidance strategies in cross-domain translation.

A. Online Map Generation Based on Deep Learning Methods

The online map is one of the productions of the digital map, reflecting various geo-information elements, such as roads, waterways, and buildings [19]. The traditional methods of cartography are based on professional experience, which are laborious and novice unfriendly. Over the past decade, the rise of deep learning has been transformative, enabling the automatic generation of online maps. The generation methods involve two paradigms: the discriminative methodology [20] and the generative methodology [21].

The discriminative methodology is considered an auxiliary tool for the online map production [22], which can supply vector data extraction. The discriminative methodology uses the convolutional neural network to extract high-level features in the RS imagery. The typical framework is the DeepLab network [23], which enhances the ability to capture spatial information via deeper layers of the network for semantic segmentation. To overcome the expensive challenges of data annotation, EasySeg network [13] employs a semi-supervised strategy to mitigate the high demands for accurate labels. Moreover, others design some discriminative networks to extract specific targets to acquire more geospatial information. For example, an automatic road extraction network was proposed [24]. However, the aforementioned methods still cannot achieve end-to-end map generation automatically.

Therefore, another mainstream method is based on the generative model, such as generative adversarial networks (GANs) [25]. The GAN-based methods regard the translation from RS images to online maps as a domain mapping task [12], [26]. However, supervised learning inevitably imposes significant data annotation burdens. Thus, Song et al. [5] propose an unsupervised method called MapGenGAN to mitigate the dependence on paired data. To further improve the generation quality, some researches focus on semi-supervised learning, which can achieve comparable performance to supervised learning approaches with a small ratio of matched data. Nevertheless, the GAN-based methods still suffer from mode collapse and less mode coverage, leading to lower fidelity and accuracy. Therefore, Tian et al. [27] employ DMs to improve the quality of map generation. Although DM-based methods cost more computational resources, the performance is remarkably superior. However, the above-mentioned methods lack explicit semantic guidance, resulting in inconsistent semantic representation between generated maps and original RS images. Thus, our model is designed to introduce progressive guidance to maintain semantic consistency.

B. Image Generation Based on DM

Since the advent of the DMs [16], the image processing task based on the DMs has gained considerable research interest. Considering controllable generation, some works proposed [28] conditional DMs to guide the direction of the denoising results. For example, ILVR [28] uses a latent variable to guide the denoising process, which achieves image translation between various domains. Latent DM (LDM) [29] offers a new paradigm, which maps images from the pixel level into the latent space via a pretrained VQ-Encoder. Based on the LDM, several works improved the image generation performance. Palette [30] is a topic model for image-to-image translation based on conditional DMs with supervised learning. Furthermore, some researchers have designed a denoising diffusion bridge model (DDBM) with fixed endpoints of the diffusion process [31]. DDBM provides a more reliable generation with no noisy randomness disturbance. Therefore, more relevant research [32], [33], [34] has used the denoising bridge framework for image translation. However, there remains room for facilitating the accuracy of semantic details in the RS imagery.

Moreover, some research employs DMs in the field of RS, demonstrating the effectiveness compared to other methods [35]. Besides, Yuan et al. [36] achieve fake sample generation based on DMs, and Wang and Sun [37] apply diffusion prior information in RS image super-resolution. These applications indicate the contributions of DMs in the field of RS. Therefore, our proposed method tries to employ progressive guidance based on the semantic prior information, providing a novel framework for effective map generation.

C. Semantic Label Guidance Strategies

Semantic label guidance offers insights into practical applications in the domain of adaptive translation. Due to the large gap across two distinct domains, many works utilize the prior labels to achieve a more convincing translation. For example, Fu et al. [12] have first combined the generative and discriminative models, introducing semantic labels into the training process. Besides, SPN2D-GAN proposed a night-to-day scenes translation network with the original gradient of the night scene images [15]. Moreover, more research focuses on the guidance strategy in the DMs to make full use of the high extensibility of the diffusion process. Peng et al. [38] proposed a progressive guidance strategy to control the generation process, achieving outstanding performance in the local details. Besides, EBDM is also designed to use mask labels in the Brownian bridge DM [39], providing a novel approach of end-to-end conditional generation. The proposed PSP-MapDiff employs semantic and edge prior information as the guidance between two adjacent denoising steps. With a slight discrepancy between the adjacent results, this progressive guidance strategy can achieve stable cross-domain translation.

III. Methodology

In this section, the architecture of our PSP-MapDiff is illustrated. Then, we introduce the details of the prior-based components of PSP-MapDiff.

A. Overall Architecture

As shown in Fig. 2, our proposed approach, PSP-MapDiff, is an end-to-end framework for online map generation, which

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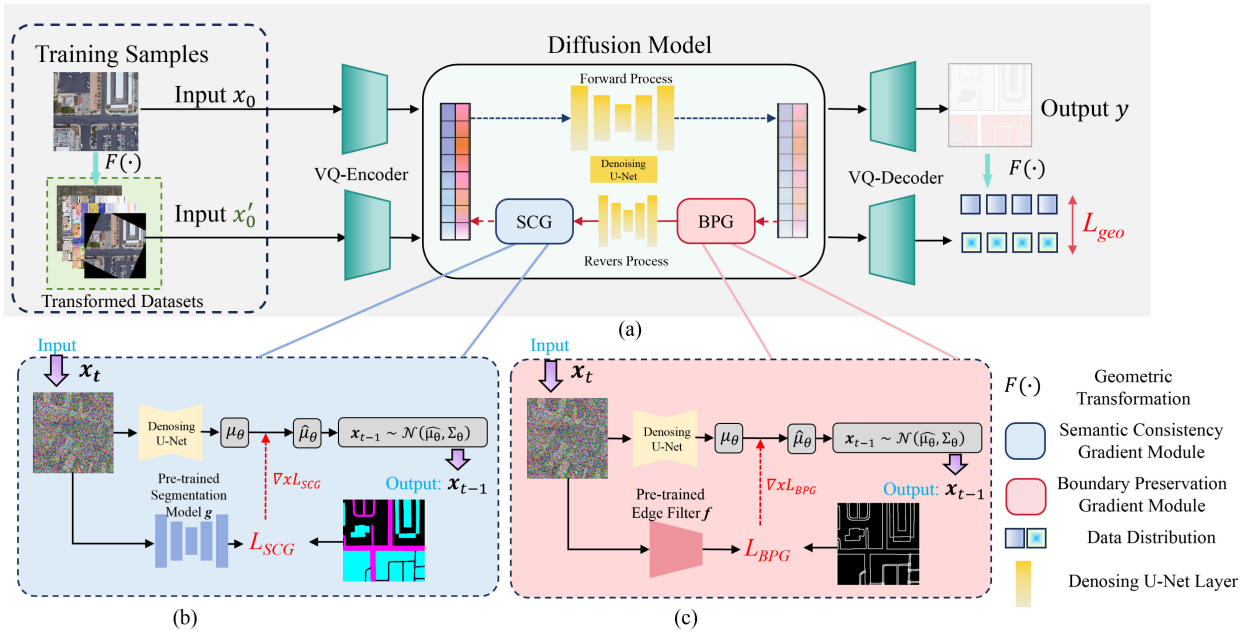


Fig. 2. Illustration of our proposed PSP-MapDiff for online map generation. (a) Pipeline of our model. The PSP-MapDiff firstly employs a VQ-Encoder to extract high-level features of RS images, and then the latent codes are trained in the DDBM with two progressive guidance modules, SCG and BPG, which can reduce the semantic bias and boundary blur. The VQ-Decoder can return the generated results to the pixel-level maps. (b) Details of the SCG module. (c) Details of the BPG module.

consists of an image compression encoder and decoder, a diffusion network, a SCG module, and a BPG module.

First, we employ a geometric transformation function $F(\cdot)$ to expand our datasets, which will be input into the VQ-Encoder together. The VQ-Encoder is based on the VQGAN model [40]. The raw data and transformed data are mapped into latent variables.

Second, we introduce a denoising U-Net with fixed endpoints, reducing the randomness caused by Gaussian noise. Prior online map generation methods lack semantic prior information guidance [27]. Thus, we design two semantic label guidance modules to mitigate the semantic error stemming from the complex conditions and enhance boundary accuracy.

Finally, through the VQ-Decoder, the results are translated from the latent codes into the pixel images and online maps. For capturing more spatial features, a geometric loss function L_{geo} is employed between the transformed data and the original data.

B. Image Compression Encoder

According to the conditional VQGAN, we introduce the conditional RGB RS image as input of VQ-Encoder in Fig. 3. In particular, given an RS image $X \in \mathbb{R}^{H\times W\times N_c}$ in the pixel space, where N_c is the number of channels, the encoder $\mathbb{E}$ translates the images into the latent variable $Z = \mathbf{E}(X) \in \mathbb{R}^{h\times w\times N_z}$. In the convolutional layers in the encoder, the downsampling factor is $f = H/h = W/w = 2^m, m \in \mathbb{N}$, and the channel of the latent variable N_z is usually larger than the channel of RS images N_c.

In order to acquire more regular variable features in the latent space, we employ a codebook $\mathcal{Z}$ to quantize the raw latent code into a paradigm variable via the following equation:

$$z_q = q(Z) = \text{argmin}_{\hat{z}_k\in\mathcal{Z}} \left|\left|z_{ij} - \hat{z}_k\right|\right| \in \mathbb{R}^{h\times w\times N_z} \quad (1)$$

where the codebook $\mathcal{Z} = \{z_k\}_{k=1}^K$ is the general quantization dictionary. We use the $L1$ distance for the quantization at each pixel at the location (i, j) of the image.

Through the DM, the CNN-Decoder can translate the generated latent features into the pixel-level online maps. The image compression module takes advantage of reducing resource costs and capturing features.

C. Semantic Prior Guidance Modules

It is well-established that DMs possess inherent generative diversity, which may produce new content deviating from RS imagery due to the substantial domain gap between RS data and map representations. To mitigate this divergence, we incorporate targeted constraints during the denoising process. Our framework introduces two critical components: SCG and BPG, which collectively enforce classification accuracy through explicit semantic guidance throughout the generation process.

In particular, we utilize pretrained discriminative models to derive class-specific segmentation masks and boundary representations from intermediate outputs. However, two loss functions are designed to evaluate segmentation results against ground-truth annotations. Then, we use the gradient mean value to constrain the denoising process.

1) Semantic Consistency Guidance, SCG: In Fig. 4, the key point of SCG is that we employ a pretrained segmentation network g, inspired by DeepLab v3+ [23], which has been modified in the classified part to adapt to our specific segmentation task. Furthermore, we design a novel distance as a loss function to guide the class classification in the progressive

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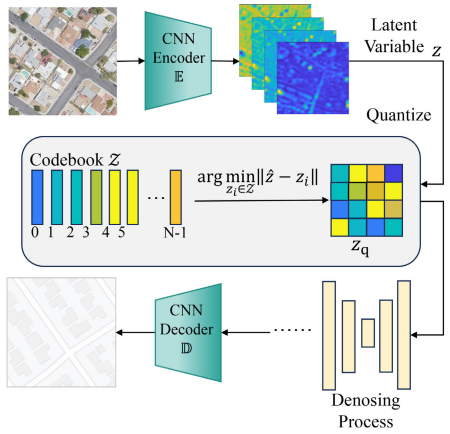


Fig. 3. Details of the VQ-Encoder and VQ-Decoder. First, we employ a CNN encoder to capture the high-level features of the pixel-level RS images. The variables Z in the latent space can be quantized by a standard codebook $\mathcal{Z}$ via $L1$ distance. After the denoising process in the diffusion U-Net, the CNN-Decoder generates the final online maps.

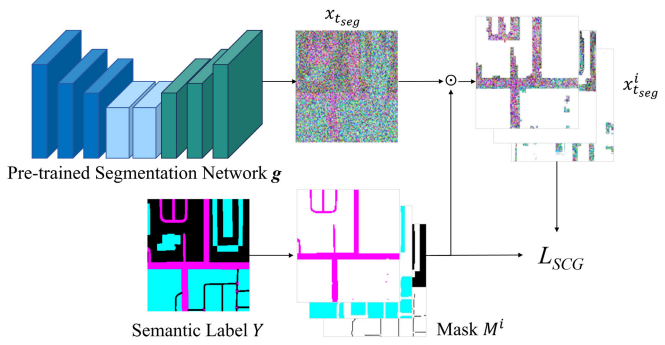


Fig. 4. Overall flowchart to calculate the L_{SCG} of the SCG module.

generation. This ensures that the semantics in the generated online maps remain consistent with the input.

The details are as follows. Given the middle result of the denoising process x_t, the segmentation network g predicts the label probabilities of the pixels x_{tseg}. Then we use a source semantic label $y \in Y$, and we can obtain the ith mask m^i, where $i \in \{0, 1, 2, 3, 4\}$, to present the entire region of the ith ground element in the RS image

$$m^i \in M = \begin{cases} 1, & \text{if pixel value in } y \text{ equals } i \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

After that, the ith prediction in the x_{tseg} is obtained by

$$x^i_{\mathrm{tseg}} = x_{\mathrm{tseg}} \odot m_i \tag{3}$$

where $\odot$ represents the Hadamard product. The x^i_{tseg} stands for the filtered prediction of the ith class based on the mask m_i.

Next, we can calculate the loss function L_{SCG} via the output of the filtered prediction and segmentation mask m_i in all regional classes

$$L_{\mathrm{SCG}} = \Sigma^K_{i=0} \mathrm{MSE}\left(x^i_{\mathrm{tseg}}, m_i\right), \quad K \in (0, 1, 2, 3, 4) \tag{4}$$

where MSE is the mean-square-error function to evaluate the distance between the true segmentation label and prediction results. The smaller the value is, the more semantic information is leveraged.

In the reverse process, given the middle denoising result x_t, the proposed denoising U-Net will estimate the average value μ_θ, and then sample from the distribution $p_\theta = \mathcal{N}(\mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$. What is more, considering the influence of the loss function L_{SCG}, the adjusted $\hat{\mu}_\theta$ is designed by the loss gradient guidance shown in Fig. 2(b)

$$\hat{\mu}_\theta(x_t, t) = \mu_\theta(x_t, t) + \alpha \Sigma_\theta \nabla_{x_t} L_{\mathrm{SCG}} \tag{5}$$

where the μ_θ is the previous mean value of the output. We argue that the middle results are influenced by the noise, which may cause error guidance. Thus, we set $\alpha = t/T$, which is a hyperparameter for gradient guidance controlling. The covariance Σ_θ is a settled value in advance. When the denoising process starts, α is small to avoid the disturbance of the random noise.

Therefore, the output of the SCG Module x_{t-1} can be sampled from the adjusted distribution followed by:

$$x_{t-1} \sim \mathcal{N}\left(\hat{\mu}_\theta(x_t, t), \Sigma_\theta\right). \tag{6}$$

2) Boundary Preservation Guidance, BPG: While the SCG utilizes prior information from semantic labels, the visual quality of the online maps also depends critically on boundary accuracy. To address this, the BPG module is proposed to enhance boundary clarity in the translated maps. Our method, inspired by the boundary loss function [41], leverages the boundary information from the dataset labels. More specifically, we implement a deep convolutional network as the edge filter. This choice allows for significantly improved accuracy in boundary detection.

Given the middle denoising result x_t, the edge filter f extracts the contours, denoted as x_{tb}. Then, we propose a boundary metric BF_1 to evaluate the precision of the boundary preservation in the generated noisy images

$$L_{\mathrm{BPG}} = 1 - \mathrm{BF}_1 \tag{7}$$

where we define the BF_1 as

$$\mathrm{BF}_1 = 2 \times \frac{p_b r_b}{p_b + r_b} \tag{8}$$

where we can utilize the filtered noisy image x_{tb} and $y \in Y_b$ to calculate the precision p_b and recall r_b, which can comprehensively evaluate the boundary quality of the generated maps.

As the same process as the SCG module, we introduce the gradient of the loss function L_{BPG} to guide the denoising direction according to (5) and (6). These two modules may offer constraints to the denoising process in an alternating manner, which we will discuss the details of the training process in Section III-E.

D. Loss Function

According to the backbone model [27], the optimizing objective in the training process can be defined as an evidence lower bound L_{ELBO}

$$\mathbb{E}_{x_0, y, \epsilon}\left[c_{\epsilon t} \left\| m_t(y - x_0) + \sqrt{\delta}\epsilon - \epsilon_\theta(x_t, t)\right)\right\|^2\Bigg] \tag{9}$$

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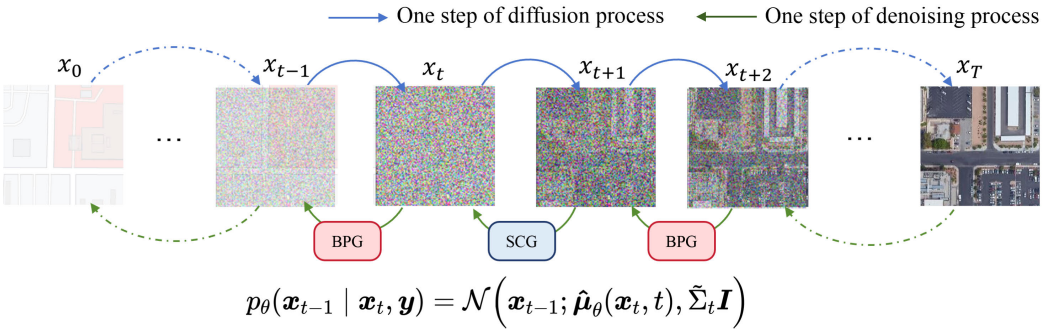


Fig. 5. Details of the progressive training strategy. We fixed the endpoint of the diffusion process x_T with the data from the RS domain. While denoising, we successively employ the SCG and BPG modules to guide the direction of generation.

where the x_t is sampled from the distribution (10). Besides, the key step is to calculate the average value $\hat{\mu}(x_t, t)$. We employ the U-Net network to predict the noise rather than calculate the whole $\hat{\mu}$ according to the (11), where the parameters c_{xt}, c_{yt}, and ϵ_t

$$q_{\mathrm{BB}}\left(x_{t-1} | x_t, x_o, y\right) = \mathcal{N}\left(x_{t-1}; \hat{\mu}\left(x_t, t\right), \Sigma_\theta\right) \tag{10}$$

$$\hat{\mu}\left(x_t, t\right) = c_{xt} x_t + c_{yt} y + c_{\epsilon t} \epsilon_t\left(x_t, t\right) + \lambda \Sigma_\theta \nabla_{x_t} L_{\mathrm{SC/BPG}}. \tag{11}$$

Therefore, the final optimizing objective can be simplified as

$$\mathbb{E}_{x_0, y, \epsilon}\left[c_{\epsilon t}\left\| m_t\left(y - x_0\right) + \sqrt{\delta}\epsilon - \epsilon_\theta\left(x_t, t\right)\right)\right\|^2\Bigg]. \tag{12}$$

Moreover, we design a geometric transformation loss function to improve the robustness and generalization of PSP-MapDiff. As shown in Fig. 2, we employ the transformation function $F(\cdot)$, like cropping, rotation, sharpening, and so on, in order to expand the input data and acquire x_0', where $x_0' = F(x_0)$

$$L_{\mathrm{geo}} = \operatorname{argmin}\left\{E_{x \sim p_{\mathrm{data}}} \left\|F\left(G\left(x_0\right)\right) - G\left(F\left(x_0\right)\right)\right\|_1\right\}. \tag{13}$$

To refine the final generation performance, we apply the sum of four kinds of loss function terms as the final loss function in PSP-MapDiff. The loss function can be formulated as (14), where λ is a hyperparameter. We argue that both semantic and edge guidance are equally detrimental to the denoising process. Therefore, we advocate 1:1 as the default weight ratio for L_{BPG} and L_{BPG}

$$L_{\mathrm{total}} = L_{\mathrm{ELBO}} + \lambda L_{\mathrm{geo}} + L_{\mathrm{SCG}} + L_{\mathrm{BPG}}. \tag{14}$$

E. Progressive Training

The key part of our PSP-MapDiff is two semantic modules, which keep the semantic consistency and boundary continuity. In order to mitigate the gap between distinct domains, we design a progressive guidance strategy, which implements gradient guidance successively in the denoising process.

Fig. 5 demonstrates the mechanism of the progressive guidance strategy with two key modules: the SCG and BPG. In the reverse process of the DM, we inject the guidance module into every gap between two adjacent denoising results. Below are implementation details.

The backbone of our model is based on [27], where the training process is divided into two stages: diffusion process and denoising process. We fix the endpoint x_T by the source domain data as shown in Fig. 5. Therefore, the sample distribution $q(x_t \sim x_T)$ can be obtained by the time-reversed stochastic differential equation (SDE). Through the Brownian motion constraint, the denoising process can be guided to a promising direction by the following equations:

$$p\left(x_t \sim x_T\right) = \mathcal{N}\left(\left(1 - \frac{t}{T}\right) x_0 + \frac{t}{T} x_T, \frac{t\left(T - t\right)}{T} I\right) \tag{15}$$

$$q_B\left(x_t \mid x_0, y\right) = \mathcal{N}\left(x_t; \left(1 - m_t\right) x_0 + m_t y, \Sigma_t I\right)$$
$$x_0 = \boldsymbol{x}, \quad m_t = \frac{t}{T} \tag{16}$$

where the T is the total diffusion step, and Σ_t is the covariance of the data distribution.

Meanwhile, we noticed that the covariance is influenced by the total step T. In order to control the value of the variance, the Σ_t can be defined below

$$\Sigma_t = 2s\left(m_t - m_t^2\right) \tag{17}$$

where s is a hyperparameter set in advance.

In the reverse process, we begin the denoising process at the RS images. Pretrained semantic segmentation model g and edge filter f implement different gradient guidance at an equal frequency. Due to the small gap between the adjacent denoising images x_t and x_{t-1}, it becomes possible for pretrained g and f models to work well in the denoising process. Fixing the parameter T, we employ each module alternately to guide the process. In addition, we apply a time-related hyperparameter $\alpha = t/T$, where T is the overall timestep. α can avoid misdirection at the denoising start point.

The proposed segmentation model g and edge filter f are well trained on source-domain labels Y_s and Y_b. Based on this operation, we can progressively implement the guidance in the reverse stage. To simply formulate this operation, we use $x_{t-1} = \phi\{x_t, g/f\}$, where $\phi\{\cdot\}$ denotes the denoising operation. Based on (5) and (6), the detail prediction of x_{t-1} can be computed by the following equation:

$$p_\theta\left(x_{t-1} \mid x_t, y\right) = \mathcal{N}\left(x_{t-1}; \hat{\mu}_\theta\left(x_t, t\right), \Sigma_\theta I\right) \tag{18}$$

where the predicted average value $\hat{\mu}_\theta$ is defined by (5) and is modeled by the neural network.

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IV. Experiments

To assess the effectiveness and generalizability of the proposed PSP-MapDiff, we compare it with several classical and advanced image-to-image translation methods and online map generation models in three well-established and widely used urban datasets. Further experimental details follow.

A. Datasets

Based on the AIDOMG datasets [12],[1] we select three representative urban areas, including Las Vegas, Cairo, and Rio de Janeiro. Their diversity and complexity (RS images, online maps, semantic labels, and edge labels) make them ideal for a comprehensive evaluation. The data examples are illustrated in Fig. 6. The resolution is 0.6 m per pixel, and the size of all images is 256×256. Meanwhile, the three datasets have distinct characteristics in terms of illumination, transparency, and ground conditions. Below are more details. The semantic labels are divided into four key categories: roads, buildings, waterways, and vegetation, by clustering the corresponding maps. Besides, we employ an edge filter to make edge labels, which contain the boundary of roads and buildings. Since prior labels only provide guidance to constrain generation, it is not necessary to label all ground objects in RS images, which means the annotation work is simpler and faster than in general semantic segmentation tasks.

1) *Las Vegas:* The RS imagery is collected from the Google Map in the middle of USA, where there is ample illumination and no cloud cover. The urban roads are well-regulated, with paved surfaces and predominantly featuring standalone buildings.
2) *Cairo:* This city is located in North Africa. The weather is dry, and the illumination varies. The city is characteristic of the dense buildings and abundant vegetation, which results in more occlusion.
3) *Rio:* The collected RS imagery features changed illumination, high road, and building density. The ground condition is more complex than in other datasets.

B. Comparison Methods

To demonstrate the performance of our proposed network, it is compared with several state-of-the-art methods for online map generation based on deep learning. These methods employ various strategies to achieve automatic map generation, including as follows.

1) Pix2pix [26] is a typical GAN-based image-to-image translation method, which is employed as a general comparison method.
2) AIDOMG [12] is the first GAN-based framework for automatic online map generation with the guidance of the discriminative model, which provides a paradigm for the combination of two kinds of models.
3) SPN2D-GAN [15] introduces the guidance of semantic labels to improve the performance of the translation model. It designs a semantic prior generator to leverage the semantic prior knowledge to enhance the fidelity of the local details.
4) Semi-MapGen [5] is a semi-supervised model for the RS image-to-online map translation. With the 50% ratio, the semi-supervised method has a better performance than the supervised baselines.
5) Stable diffusion model (SDM) [29] is a typical DM for the image generation task, which introduces the attention mechanism for diffusion/denoising process in the latent space.
6) I2SB [32] proposes Schrödinger Bridge, a nonlinear diffusion framework for the image-to-image translation task.
7) UNSB [33] proposes Unpaired Neural Schrödinger Bridge with unpaired data, allowing it to incorporate with the discriminators and regularization.
8) MapGen-Diff [27] is the first exploration for online map generation based on the DM, which achieves good performance in two widely used datasets.

To evaluate the performance of the methods, four metrics are employed for the quantitative assessment of generation. Considering the tradeoff between subjective creativity and objective reality, we employ root-mean-square error (RMSE), structural similarity index measure (SSIM), and accuracy (ACC) to objectively evaluate the distance between the generated maps and truth. Besides, the learned perceptual image patch similarity (LPIPS) is employed to measure the similarity, which is more in line with human perception

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2} \tag{19}$$

$$\mathrm{SSIM}(\hat{y}, y) = \frac{\left(2\mu_{\hat{y}}\mu_y + C_1\right)\left(2\delta_{\hat{y}y} + C_2\right)}{\left(\mu_{\hat{y}}^2 + \mu_y^2 + C_1\right)\left(\delta_{\hat{y}}^2 + \delta_y^2 + C_2\right)} \tag{20}$$

$$\mathrm{ACC} = \begin{cases} 1, & \text{if } \max\{\mathrm{dis}(r_i, g_i, b_i)\} < C \\ 0, & \text{otherwise} \end{cases} \tag{21}$$

$$\mathrm{LPIPS}(y, y_0) = \sum_l \frac{1}{H_l W_l}\sum_{h,w}\left|\left|w_l \odot \left(\hat{y}^l_{hw} - \hat{y}^l_{0hw}\right)\right|\right|_2^2 \tag{22}$$

$$\mathrm{mIoU} = \frac{1}{N}\sum_{i=1}^{N}\mathrm{IoU}_i = \frac{1}{N}\sum_{i=1}^{N}\frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FP}_i + \mathrm{FN}_i} \tag{23}$$

where the y_i and $\hat{y}_i$ are the generated results and ground truth, and the μ and δ are the corresponding average and variance. C is the threshold, which is set to 5 according to [26]. The r_i, g_i, and b_i are the RGB values of the ith pixel. In (23), H and W are the sizes of the channel dimension at the lth feature stack, and $L2$ distance can be calculated.

C. Implementation Details

For all experiments in this study, the initial learning rate is set to 0.0002 with a weight decay of 0.0001. However, the optimizer uses Adam with the momentum set as 0.9 and 0.99. In the DM, we set the diffusion step to 1000, and the denoising step T is 200. The covariance Σ_θ is fixed by the $s = 0.5$. Meanwhile, the downsampling factor $f = 4$ of the image compression module. The pretrained segmentation network

[1] https://github.com/ying-fu/Aerial-Map-Translation

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TABLE I

METRIC COMPARISON OF DIFFERENT METHODS ON THE VEGAS DATASET. THE UPWARD ARROW INDICATES THAT THE HIGHER THE SCORE, THE BETTER THE MODEL'S PERFORMANCE ON THIS METRIC. (THE HIGHEST SCORE MARKED IN BOLD FONT; RUNNER-UP UNDERLINED)

Methods	RMSE ↓	SSIM ↑	ACC(%) ↑	LPIPS ↓	mIoU ↑
pix2pix [26]	20.0823	0.7615	44.38	0.350	45.84
AIDOMG [12]	<u>19.0089</u>	0.7927	<u>57.87</u>	<u>0.230</u>	51.50
SPD2N-GAN [15]	20.2008	0.7738	48.40	0.286	<u>56.25</u>
Semi-MapGen [5]	20.6073	0.7590	34.34	0.325	49.27
SDM [29]	24.7259	0.7845	45.26	0.450	47.20
I2SB [32]	19.3695	<u>0.8003</u>	46.36	0.363	47.48
UNSB [33]	25.6306	0.7198	41.62	0.386	43.20
MapGen-Diff [27]	19.4199	0.7979	53.63	0.248	47.32
PSP-MapDiff	**17.2522**	**0.8357**	**62.68**	**0.191**	**59.74**

is based on the Deeplab v3+ framework [23], and the edge filter is based on the deep convolutional neural networks. The pretrained discriminative models are trained on 500 pairs of labels. We set the hyperparameter $\lambda = 10$ in the loss function. We carried out all of the training on a PC with an Intel Core[2] i7-13700H CPU @ 2.40 GHz, 16-GB RAM, and a GeForce GTX 4090 GPU.

D. Experimental Results

We conduct experiments on the Vegas, Cairo, and Rio datasets to validate the effectiveness of PSP-MapDiff through quantitative metrics and visual comparison. The comprehensive experiments provide a sufficient evaluation of our framework.

1) Evaluation and Analysis on Vegas: The metric scores on Vegas are represented in Table I. We calculate and summarize RMSE, SSIM, ACC, and LPIPS metrics based on Vegas.

From Table I, it can be seen that the proposed semantic guidance model PSP-MapDiff performs better than these state-of-the-art approaches in all metrics on the Vegas dataset. The PSP-MapDiff gets the highest RMSE, SSIM, ACC, and LPIPS scores on Vegas. Meanwhile, the AIDOMG also acquires good results in most metrics. Compared to the runner-up methods, the proposed method acquires the 9.2%, 4.4%, 8.3%, and 16.9% improvements in visual quality metrics, which can be attributed to a deeper integration between the two kinds of models in the diffusion process. In addition, the proposed model achieves 5.8% enhancements in the semantic consistency metric, mean intersection over union (mIoU). It can prove the effectiveness of the guidance of prior semantic information.

Furthermore, on the Vegas dataset, the boundaries of roads and buildings are clear, which means that the clean data enables the GAN-based model to train effectively. On the contrary, due to the limitation of the noisy disturbance, the baseline methods, like UNSB, achieve the lowest scores in the quantitative evaluation. Especially, in the RMSE and ACC, we calculate the pixel-level accuracy to evaluate the generation quality. Although DM-based methods achieve good performance in visual quality reflected in SSIM scores, the RMSE and ACC still contain a large gap compared with our method.

To illustrate more evaluation virtually, we represent selected generated maps of different methods in Fig. 7. From Fig. 7, we can observe that PSP-MapDiff has more advantages in the semantic classification in the complex urban area. For example, in the third column, the urban vegetation and streets are interwoven, which makes it challenging to classify correctly. Owing to the semantic guidance, our method can recognize the vegetation correctly and stay close to the scene of the original RS images.

2) Evaluation and Analysis on Cairo: In the Cairo dataset, the PSP-MapDiff model achieves better performance, as evidenced by the quantitative metrics in Table II. More specifically, our model achieves 2.5% and 18.1% improvements over the suboptimal method AIDOMG in RMSE and LPIPS, and makes 4.6% and 14.7% improvements in SSIM and ACC against MapGen-Diff.

Moreover, the Cairo dataset consists of more comprehensive conditions in the RS imagery, resulting in decreasing scores of the evaluation metrics. Especially, for models without semantic guidance, such as pix2pix, Semi-MapGen, I2SB, and UNSB, the coverage and shadow may result in semantic errors and blurred boundaries of ground objects. In the RMSE and ACC, the SDM shows subpar performance, which indicates that noise disturbances, like illumination changes, have a huge influence on the model's performance.

For more visualization comparison, Fig. 8 illustrates the samples of generated maps. In the fourth and sixth columns, there are shadow blockings on the roads marked by the red rectangle. Our proposed method has superior performance in maintaining consistency at the shadow location. Besides, the PS-MapDiff enables generating clearer segmentation edges than other baseline methods.

3) Evaluation and Analysis on Rio: Table III and Fig. 9 present comprehensive performance comparisons of multiple

[2]Trademarked.

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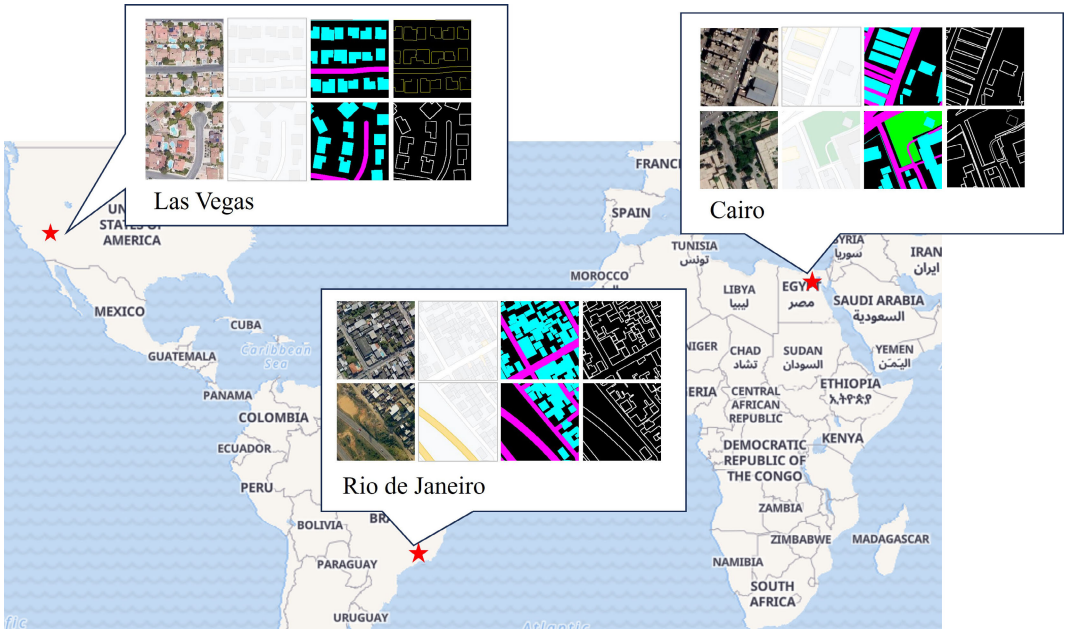


Fig. 6. Details of the SSM. Flexible thresholds are determined according to the distribution across different categories in the histogram, where the class with fewer samples is assigned a lower threshold. Compared to the fixed threshold δ_0, the dynamic thresholds $\delta(c)$ can accommodate more samples for minority classes.

TABLE II

METRIC COMPARISON OF DIFFERENT METHODS ON THE CAIRO DATASET. THE UPWARD ARROW INDICATES THAT THE HIGHER THE SCORE, THE BETTER THE MODEL'S PERFORMANCE ON THIS METRIC. (THE HIGHEST SCORE MARKED IN BOLD FONT; RUNNER-UP UNDERLINED)

Methods	RMSE ↓	SSIM ↑	ACC(%) ↑	LPIPS ↓	mIoU ↑
pix2pix [26]	25.2195	0.6917	36.31	0.439	60.15
AIDOMG [12]	<u>21.7760</u>	0.7422	52.82	<u>0.288</u>	<u>67.53</u>
SPD2N-GAN [15]	26.1200	0.7044	40.73	0.366	62.67
Semi-MapGen [5]	21.9723	0.7446	44.47	0.346	57.29
SDM [29]	25.6727	0.7430	44.51	0.386	61.35
I2SB [32]	27.2488	0.7053	36.82	0.363	50.34
UNSB [33]	25.7823	0.7322	43.31	0.395	4.078
MapGen-Diff [27]	24.0341	<u>0.7696</u>	<u>50.30</u>	0.307	66.64
PSP-MapDiff	**21.2195**	**0.8052**	**57.71**	**0.236**	**71.32**

state-of-the-art methods on the Rio dataset, evaluated through four metrics critical for semantic map generation tasks. Our proposed PSP-MapDiff demonstrates superior performance, achieving the best scores in RMSE 19.4507, SSIM 0.6725, ACC 48.70%, and LPIPS 0.198, while attaining comparable excellence in mIoU 61.36.

The experimental results reveal that PSP-MapDiff surpasses baseline frameworks across all metrics. Traditional GAN-based methods like pix2pix exhibit the weakest performance, limited by their inability to capture fine-grained structural details. AIDOMG, while achieving the second-best RMSE (19.8409), falters in SSIM and ACC, suggesting an imbalance between pixel-level precision and high-level semantic consistency. SPD2N-GAN, though notable for its suboptimal SSIM (0.6597) and ACC (48.48%), underperforms significantly in RMSE (19.9612), reflecting instability in geometric fidelity. Modern diffusion-based approaches (e.g., SDM and MapGen-Diff) show moderate improvement over traditional models but still lag behind PSP-MapDiff in key metrics—a 7.6% improvement in RMSE and 7.9% increase in SSIM compared to SDM highlight our method's robustness. Despite their well-designed pipelines, progressive denoising frameworks such as I2SB and UNSB still exhibit significant semantic distortion, as indicated by their poor LPIPS scores (0.436 and 0.386, respectively) and low ACC (less than 40%).

Notably, PSP-MapDiff's LPIPS is similar to AIDOMG, confirming its perceptual realism. This balanced advancement across complementary metrics underscores PSP-MapDiff's suitability for high-precision geospatial applications requiring both quantitative accuracy and visual coherence.

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Fig. 7. Visualization Comparison of the different methods on the Vegas dataset. (a) Input. (b) Ground truth. (c) Ours. (d) Pix2pix. (e) SPN2D-GAN. (f) AIDOMG. (g) Semi-MapGen. (h) SDM. (i) I2SB. (j) UNSB. (k) MapGen-Diff. The red box highlights the improved performance of the proposed model in the qualitative results.

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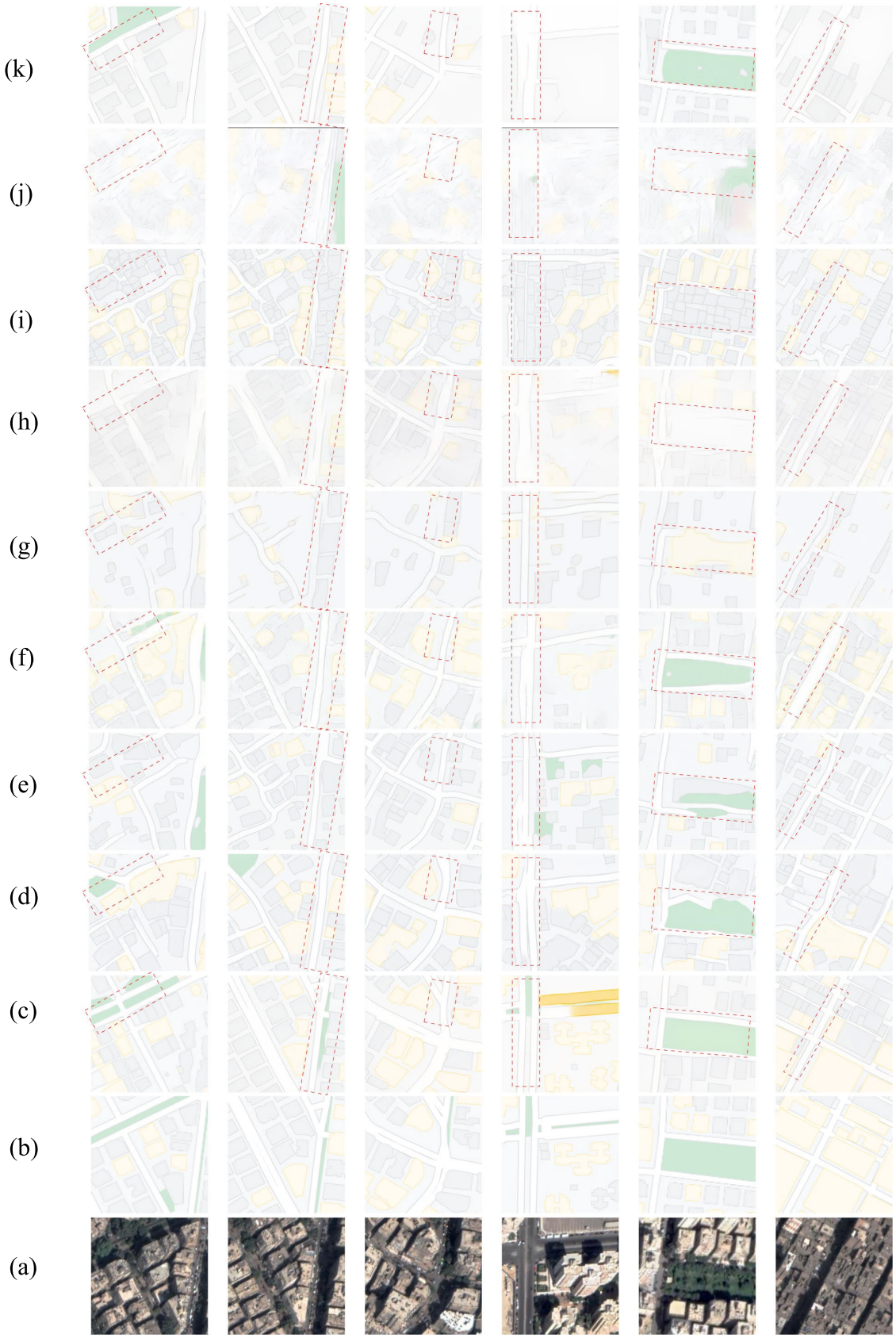


Fig. 8. Visualization comparison of the different methods on the Cairo dataset. (a) Input. (b) Ground truth. (c) Ours. (d) Pix2pix. (e) SPN2D-GAN. (f) AIDOMG. (g) Semi-MapGen. (h) SDM. (i) I2SB. (j) UNSB. (k) MapGen-Diff. The red box highlights the improved performance of the proposed model in the qualitative results.

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TABLE III

METRIC COMPARISON OF DIFFERENT METHODS ON THE RIO DATASET. THE UPWARD ARROW INDICATES THAT THE HIGHER THE SCORE, THE BETTER THE MODEL'S PERFORMANCE ON THIS METRIC. (THE HIGHEST SCORE MARKED IN BOLD FONT; RUNNER-UP UNDERLINED)

Methods	RMSE ↓	SSIM ↑	ACC(%) ↑	LPIPS ↓	mIoU ↑
pix2pix [26]	22.5843	0.5808	35.30	0.308	56.26
AIDOMG [12]	<u>19.8409</u>	0.6008	44.91	<u>0.203</u>	<u>60.05</u>
SPD2N-GAN [15]	19.9612	<u>0.6597</u>	<u>48.48</u>	0.224	56.91
Semi-MapGen [5]	23.1698	0.5697	42.52	0.219	54.71
SDM [29]	21.0555	0.6234	45.92	0.238	52.59
I2SB [32]	39.5167	0.5011	39.15	0.436	55.36
UNSB [33]	37.0259	0.5358	36.86	0.386	49.85
MapGen-Diff [27]	21.2679	0.6208	46.77	0.242	50.03
PSP-MapDiff	**19.4507**	**0.6725**	**48.70**	**0.198**	**61.36**

TABLE IV

ABLATION STUDY RESULTS BY REMOVING DIFFERENT PARTS OF THE MODEL ON THE VEGAS DATASETS

Methods	Vegas Datasets				
	RMSE ↓	SSIM ↑	ACC(%) ↑	LPIPS ↓	mIoU ↑
No-L_{SCG}	19.4606	0.8076	53.91	0.252	49.57
No-L_{BPG}	18.7084	0.7874	54.87	0.254	43.93
No-L_{geo}	17.0693	0.8336	60.13	0.206	47.31
With-f=2	18.9165	0.8219	55.41	0.285	50.15
With-f=8	20.2445	0.7890	47.42	0.292	49.64
With-f=16	20.2686	0.8076	40.16	0.331	46.22
ours	**17.2522**	**0.8357**	**62.68**	**0.191**	**59.74**

E. Ablation Study

In order to validate the effect of different components of the proposed model, such as the SCG module, the BPG module, and the geometric transformation loss, we conduct some ablation study experiments, as described below.

1) *No-L_{SCG}:* We remove the SCG module in the denoising process, where the only guidance is from the edge labels.
2) *No-L_{BPG}:* We remove the BPG module in the denoising process, where the only guidance is from the semantic labels.
3) *No-L_{geo}:* To validate the effectiveness of transformation on the data, we remove the geometric transformation loss function in the ablation study.
4) *With-f = 2:* We set the hyperparameter f to 2 so that the images are not compressed into the latent space.
5) *With-f = 8:* We set the hyperparameter f to 8 so that we can explore the influence of different levels of the latent variable.
6) *With-f = 16:* We set the hyperparameter f to 16 so that we can explore the influence of different levels of the latent variable.

Tables IV–VI demonstrate the ablation study results of PSP-MapDiff on the Vegas, Cairo, and Rio datasets, respectively. First, we assess the influence of removing different modules. When removing the SCG and BPG modules, all metric scores decrease, especially the ACC. The accuracy values decrease by 14% and 5% on the Cairo and Rio datasets, which means that the semantic prior information can enhance the model's capability to recognize ground elements with different semantic classifications. Especially, we utilize the mIoU to verify both the generated quality and the preservation of semantic information. We utilize the pretrained segmentation networks, where the input data are the generated maps. When the value of mIoU is higher, more information is preserved. We find that the mIoU decreases by 13.8% and 33.5% on the Cairo and Rio datasets, respectively, which indicates that the SCG module is beneficial for reducing semantic chaos in the generated maps. As for the other modules, it is evident that the BPG and L_{geo} can improve the performance of the proposed model. Besides, for different conditions in the three datasets, the individual modules of the model exhibit varying levels of contribution to performance. For example, the BPG module enables PSP-MapDiff to achieve 15% and 3% improvements in the RMSE on the Cairo and Rio datasets, where there are more complex scenarios.

In addition, we conduct experiments with different values of the downsampling factor f. We set f to values of 2, 4 (default), 8, and 16. From the quantitative scores in Tables IV–VI, when the factor f is set to 8 and 16, the metric scores have decreased. Especially, the pixel accuracy (ACC) metric reflects that the pixel-level values are sensitive to the hyperparameter. We argue that a higher downsampling factor may result in uncontrollable diversity due to high-dimensional features being input into the generation process. To verify the sensitivity of the proposed model to the factor f, we demonstrate the visualization results in Fig. 10.

Fig. 10 illustrates the qualitative results of ablation experiments on Vegas, Cairo, and Rio. We employ the pretrained segmentation network to acquire the semantic maps and produce error maps marked in the original RS images, which

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Fig. 9. Visualization comparison of the different methods on the Rio dataset. (a) Input. (b) Ground truth. (c) Ours. (d) Pix2pix. (e) SPN2D-GAN. (f) AIDOMG. (g) Semi-MapGen. (h) SDM. (i) I2SB. (j) UNSB. (k) MapGen-Diff. The red box highlights the improved performance of the proposed model in the qualitative results.

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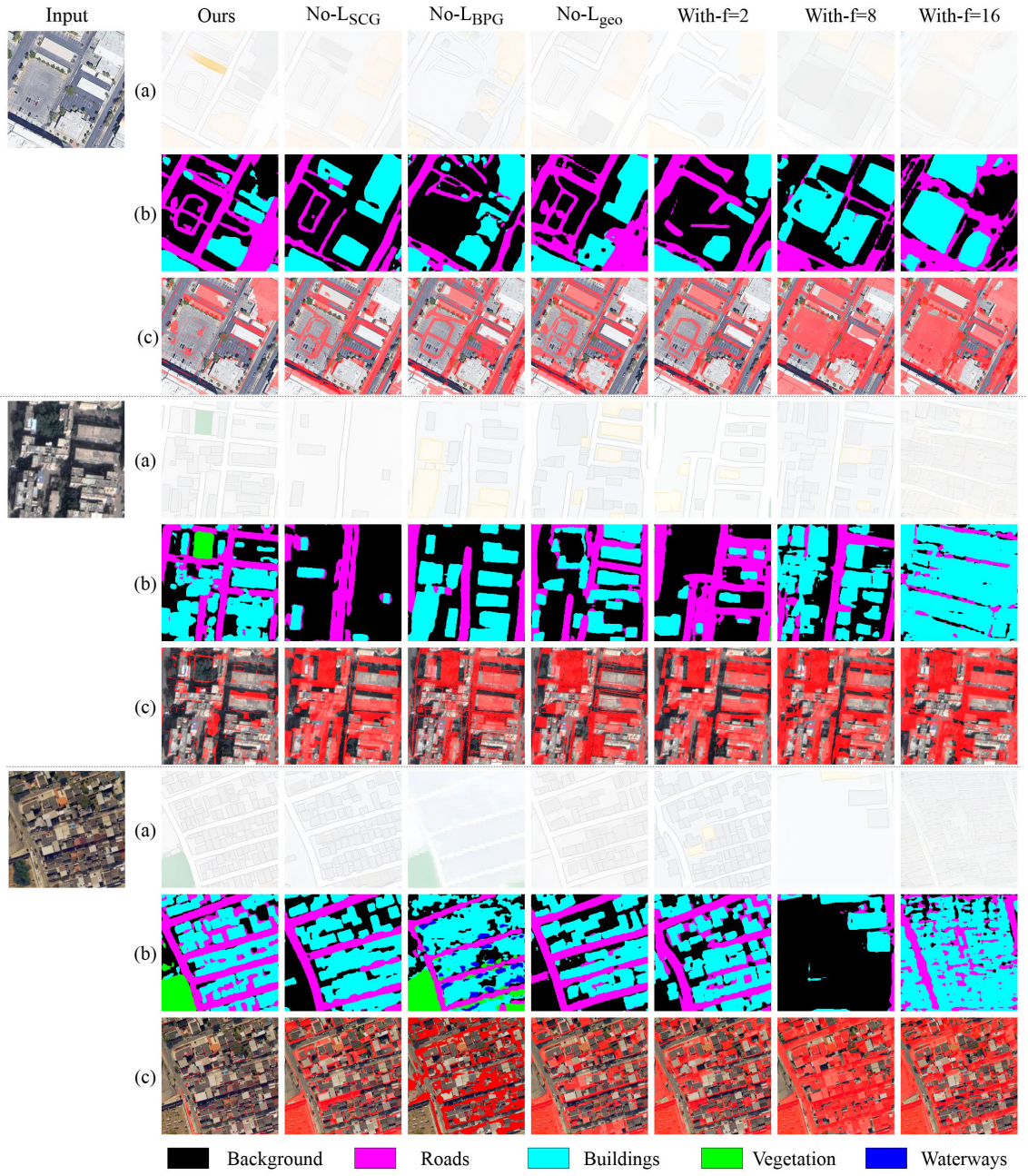


Fig. 10. Visualization of the ablation study on three datasets. (a) Generated online maps. (b) Segmentation results by the pretrained segmentation network. (c) Error prediction maps. The red regions mean the error generation, where we compare the generated results with the truth label by pixel values. The red marks mean semantic misclassification. From top to bottom, there are examples of Vegas, Cairo, and Rio datasets in order.

can directly show the classification accuracy of the generated maps. Therefore, it can be observed that our full model outperforms other ablation baselines in road and building recognition. Notably, in the Cairo and Rio datasets, there are more shadow areas where the model is more susceptible to being misled and generating incorrect maps. From the results, our model overcomes this challenge with fewer error marks, and the overall visual quality is acceptable. On the contrary, removing the SCG and BPG modules, the volume of error areas increases a lot, which confirms that the SCG and BPG modules effectively capture more prior knowledge of the semantic information. The different values of the f also impact the performance of our model. For example, the higher downsampling factor may result in uncontrollable generation due to the abstract features.

To sum up, our proposed PSP-MapDiff employs the image compression module to acquire stable and effective training in the feature space. What is more, the semantic prior guidance modules, SCG and BPG, learn to enhance the local detail generation by making full use of the prior information. The geometric transformation loss function improves the robustness and generalization of the model.

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TABLE V
ABLATION STUDY RESULTS BY REMOVING DIFFERENT PARTS OF THE MODEL ON THE CAIRO DATASETS

Methods	Cairo Datasets				
	RMSE ↓	SSIM ↑	ACC(%) ↑	LPIPS ↓	mIoU ↑
No-L_{SCG}	23.6078	0.7780	50.55	0.310	61.45
No-L_{BPG}	25.0841	0.7072	43.59	0.341	68.47
No-L_{geo}	23.3708	0.7839	52.74	0.283	64.30
With-f=2	25.1368	0.7131	45.15	0.385	62.91
With-f=8	24.5079	0.7315	48.56	0.309	56.95
With-f=16	23.6057	0.7663	32.51	0.340	50.31
ours	**21.2195**	**0.8052**	**57.71**	**0.236**	**71.32**

TABLE VI
ABLATION STUDY RESULTS BY REMOVING DIFFERENT PARTS OF THE MODEL ON THE RIO DATASETS

Methods	Rio Datasets				
	RMSE ↓	SSIM ↑	ACC(%) ↑	LPIPS ↓	mIoU ↑
No-L_{SCG}	20.6499	0.6390	43.88	0.248	40.81
No-L_{BPG}	20.0300	0.5989	48.79	0.203	54.98
No-L_{geo}	20.2884	0.6374	44.63	0.227	47.55
With-f=2	20.1672	0.6061	40.65	0.271	54.11
With-f=8	23.3877	0.5836	31.49	0.361	52.78
With-f=16	21.3036	0.5881	31.80	0.336	23.66
ours	**19.4507**	**0.6725**	**48.70**	**0.198**	**61.36**

V. DISCUSSION

A. Effect of the Semantic Prior Information Guidance

It is acknowledged that our progressively guiding strategy achieves good performance in semantic accuracy and visual quality, which is worthy of more discussion.

1) Pretrained Models and Data Quality: Compared to the typical architecture of DMs, we design a progressive semantic prior information guidance strategy to mitigate semantic errors and edge distortion by the pretrained segmentation networks. To assess the sensitivity to the different kinds of discriminative models, we test other semantic segmentation models and edge detection filters. As the default, we select the DeepLab v3+ [23] and DeepEdge [42] as the pretrained models. We also test other segmentation models and traditional edge filters.

According to the survey [43], DeepLab v3+ has superior performance compared to other networks. From Table VII, the quantitative scores have dipped when replacing the segmentation network with DeepUNet and UNet, confirming that the performance of generation is sensitive to the performance of the segmentation models. Similarly, when employing simple edge detection methods, like Sobel and Canny, the metric scores also decrease. In fact, PSP-MapDiff is more like a general framework since it can incorporate different existing semantic discriminative models. Accordingly, the superior accuracy of new semantic discriminative models can enhance our model's performance in generating online maps.

TABLE VII
COMPARISON OF DIFFERENT SEMANTIC SEGMENTATION NETWORKS AND EDGE FILTERS IN QUANTITATIVE METRICS

	Methods	RMSE ↓	SSIM ↑	ACC(%) ↑	LPIPS ↓
Segmentation Model	DeepLab v3+	**17.2522**	0.8357	**62.68**	**0.191**
	DeepUNet	18.6172	**0.8450**	61.26	0.225
	UNet	20.8911	0.7855	60.82	0.295
Edge Filter	DeepEdge	**17.2522**	**0.8357**	**62.68**	**0.191**
	Sobel	24.9179	0.6122	47.01	0.254
	Canny	22.3502	0.7281	46.98	0.271

In addition to the performance of discriminative models, the quality of prior labels also significantly influences the effectiveness of map generation. It is also observed that with a limited number of training data, the discriminative model can provide effective guidance and help the generative model achieve high-quality generation. However, the discriminative model relies on the quality of training data, which still brings the risk of misguidance from the accumulated bias.

2) Visualization of the Embedding Features: To further illustrate the effect of the SCG and BPG modules, Fig. 11 employs the t-SNE method, embedding the high-level features distribution of the network with and without the model's components. The high-level features are extracted from the last convolution layer, where we can visually observe the class clusters of the segmentation results. Compared to the features from the model with the SCG module, there are more mixed feature dots in the roads and buildings feature clusters without the SCG module. Besides, removing the BPG module, the boundary of the building and road feature clusters is mixed up. The buildings and roads are the most common objects in the RS images. The BPG module can make the feature distribution of roads and buildings more discriminative. Therefore, under the guidance of the semantic prior guidance modules, our model can extract more discriminative features in the latent space.

B. Limitations

Our framework is based on the DDBM, which needs more computational resources. Therefore, we have to declare some limitations of the proposed methods for specific applications.

1) Complexity Analysis of Models: Although the proposed model achieves outstanding performance in generation metrics, the complexity is also a key metric to evaluate an algorithm. Thus, we analyze computing complexity in this section, including time complexity for assessing the computing efficiency and space complexity for auxiliary space requirements. From Table VIII, GAN-based models demonstrate superior computational efficiency with a significantly reduced number of parameters and inference time. While our approach achieves superior performance in semantic consistency, it requires extended inference time to accomplish the progressive guidance process. It must be acknowledged that the generation

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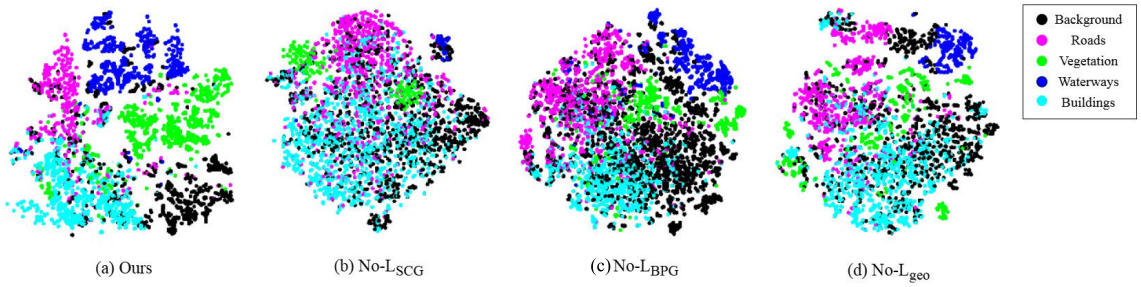


Fig. 11. Visualization of high-dimensional feature distribution applying t-SNE with different components. (a) Embedding feature distribution of PSP-MapDiff. (b) Feature distribution with removal of the SCG module. (c) Feature distribution with removal of the BPG module. (d) Feature distribution with removal of the geometric transformation loss.

TABLE VIII
COMPARISON OF THE COMPLEXITY WITH OTHER BASELINES AND INFERENCE TIME PER 256 × 256 IMAGE

	Methods	Parameters	FLOPs	Time
GAN-based	pix2pix	54M	18.15G	0.01s
	AIDOMG	55M	-	0.01s
	SPD2N-GAN	146M	94G	0.02s
	Semi-MapGen	218M	92.35 G	0.15s
DM-based	SDM	270M	200G	2.00s
	I2SB	730M	448G	48.0s
	UNSB	80M	-	0.05s
	MapGen-Diff	237M	169G	9.00s
	PSP-MapDiff	332M	150G	7.00s

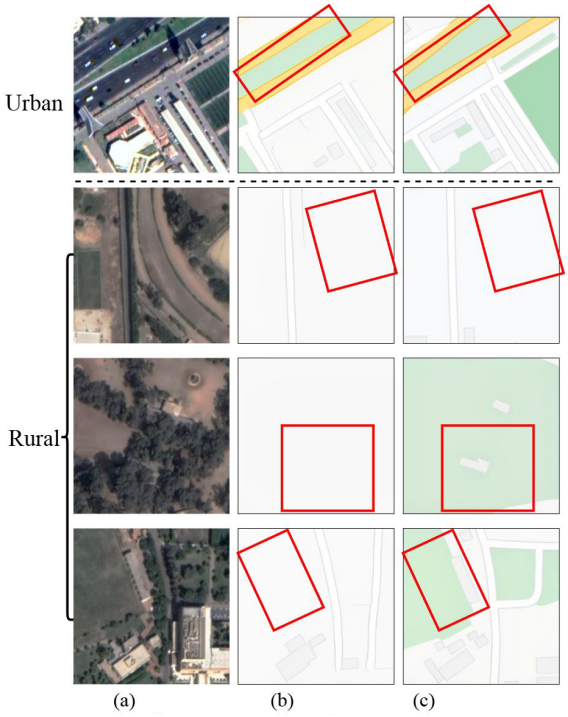


Fig. 12. Generalization visualization in different areas. The upper is from an urban street. The one is from rural areas with low illumination. (a) Input. (b) Output. (c) Ground truth.

speed of the proposed model imposes limitations for real-time applications. Consequently, subsequent research will focus on acceleration strategies and addressing the tradeoff between generation quality and computational efficiency.

2) Evaluation of Generalization: We confirm that online map generation in urban areas can bring higher economic and social value. So that we select three urban datasets with different styles to validate our model performance. Nevertheless, to comprehensively assess the model's generalization capability and broaden its applicability, we systematically incorporated rural datasets for extended analysis.

Fig. 12 illustrates generated maps from both urban and rural regions. Notably, urban scenes exhibit more structured spatial distributions, whereas rural images are frequently affected by suboptimal illumination, potentially degrading model performance. For example, the red rectangle highlights how urban green spaces—systematically integrated between buildings and roads—are readily identified by the network. In contrast, rural areas suffer from reduced generation quality due to unfavorable lighting conditions.

In addition, dynamic environmental factors such as seasonal variations limit the model's generalization in nonurban landscapes, including agricultural and forested regions.

VI. CONCLUSION

In this work, we propose a DM, named PSP-MapDiff, combined with the segmentation networks for the online map generation from the RS imagery. Our framework emphasizes the efficient incorporation of the discriminative model and the generative model, which can achieve online map generation with inference creativity and semantic consistency. We design a progressive guidance strategy with semantic prior information to bridge the domain gap between realistic RS images and abstract maps. In particular, PSP-MapDiff comprises two core modules, the SCG module and the BPG module, which can effectively leverage the semantic prior information. Furthermore, the geographic transformation loss enhances the robustness of our model. The extensive experiments reveal that the proposed model gains outstanding performance compared to the state-of-the-art methods. Therefore, the proposed framework provides a new paradigm for automated cartography,

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supporting more applications of urban planning and disaster response.

In future work, we will pay more attention to the usage and incorporation of foundational models with multimodal data, like SAR images and GIS data. We would also like to focus on generating more conditions on the 2-D and 3-D data, in order to achieve more generalized applications.

ACKNOWLEDGMENT

The authors would like to thank the anonymous referees for their valuable comments and helpful suggestions.

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